

Patient Information

Patient ID	RTA-2417	Name	Test1
Age / Sex	4 yrs / M	Diabetes	4 years duration
Previous DR	None	HbA1c	—
Prev. Screening	—	Screening Date	20 Sep 2026

Image Quality Assessment

Eye	Status	Focus	Brightness	Contrast	FOV	Enhanced
Left	GOOD	419	59	100	97	No
Right	—	—	—	—	—	—

AI Screening Results

LEFT EYE
Grade 1 Mild NPDR
NON-REFERABLE
Image Quality: Good

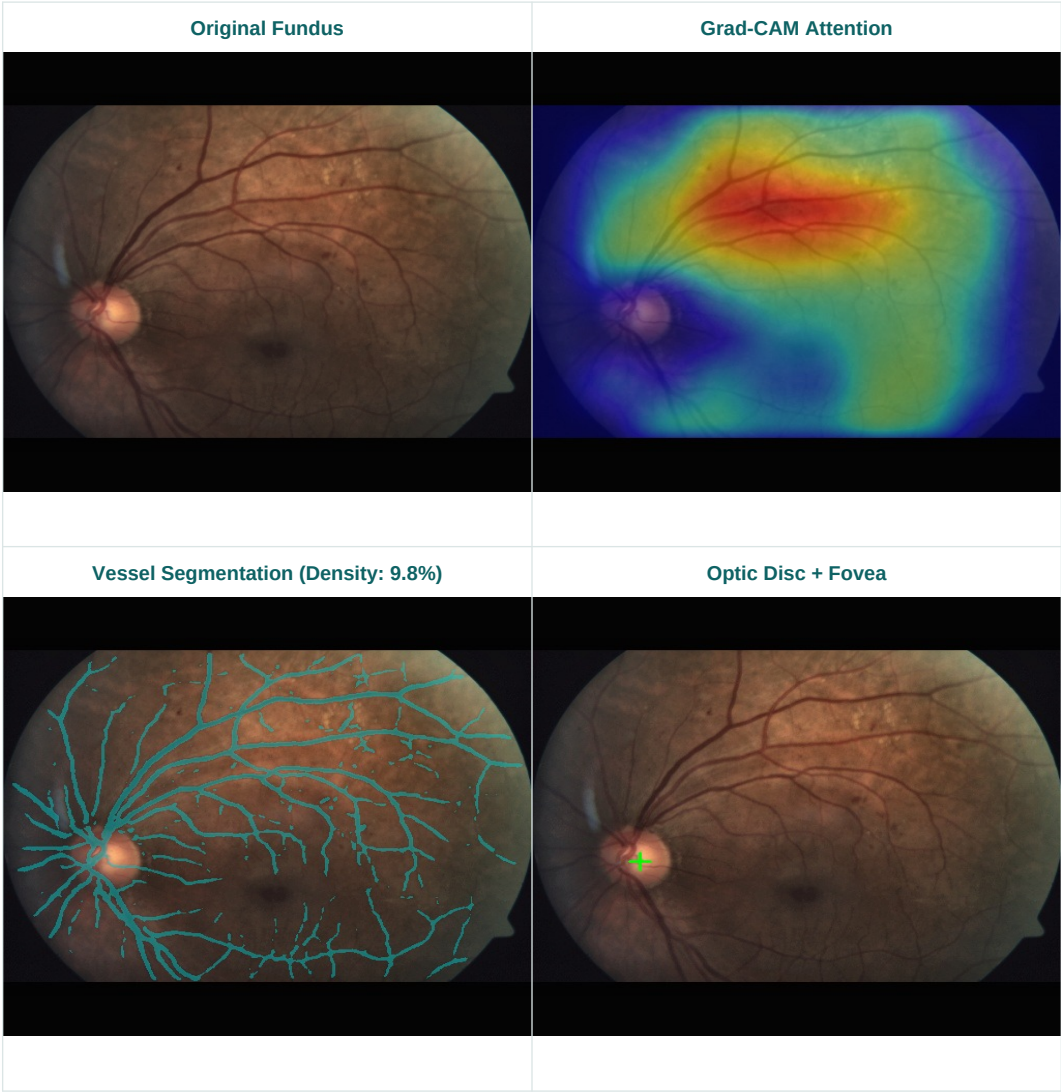
RIGHT EYE
— Not analysed
—
Image Quality: N/A

Left Eye — Grade Probability Distribution

Grade 0 — No DR		10.5%
Grade 1 — Mild		80.4%
Grade 2 — Moderate		2.7%
Grade 3 — Severe		6.2%
Grade 4 — Proliferative		0.3%

Visual Evidence

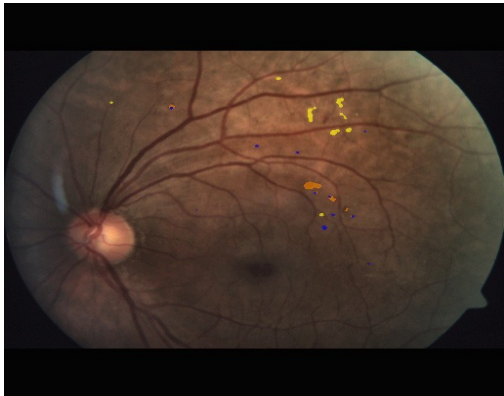
Left Eye — Retinal Analysis



Lesion Analysis & Microvascular Pathology

Left Eye — Microvascular Lesion Breakdown

Left Eye · Lesion Overlay



Lesion Type	Map Color	Status	Count
Microaneurysms	Blue	Detected	11
Intraretinal Hemorrhages	Orange	Detected	7
Hard Exudates	Yellow	Detected	9
Neovascularization	Clear	Not detected	0

● **High microvascular lesion load**

Total 27 focal lesions quantified via 5-fold ensemble.

• **Macular Advisory:** Hard exudates present — evaluate foveal proximity for DME risk.

Key: ■ Microaneurysms ■ Hemorrhages ■ Exudates

Retinal Vascular Biomarkers — Research / Exploratory Morphometry

Retinal Microvascular Biomarkers (RMB) Summary			
Arteriolar-to-Venular Ratio (AVR)	0.1118	—	Parr-Hubbard-Knudtson (Zone B Heuristic)
CRAE / CRVE Calibers	4.1 / 37.0 px	—	Central Retinal Equivalents (px)
Mean Distance Tortuosity (τ_d)	0.0159	—	Arc / Chord - 1 (Skeleton branches)
Curvature Tortuosity (τ_c)	0.0271	—	Mean angular deviation / arc length
Fractal Dimension (D_f)	1.3104 ($R^2=0.980$)	—	Box-Counting complexity & R^2 fit
Branching Topology	22 zb, 102 br	—	Zone-B vessels / Bifurcations

* Retinal microvascular biomarkers (AVR, tortuosity, fractal dimension) are computed algorithmically via MATLAB Engine morphometry and heuristics. They are intended for investigative research and do not constitute standalone clinically validated diagnostic measurements.

Referral Recommendation

✓ ROUTINE FOLLOW-UP

Routine annual screening recommended. No immediate referral indicated.

AI pipeline completed in 32.2 seconds.

DISCLAIMER: This report is generated by the RetinaAI AI-assisted screening system (SIH 26038) and is intended for use as a clinical decision-support tool only. AI outputs are not a substitute for professional medical examination. Final clinical decisions must be made by a qualified healthcare professional. Grad-CAM attention maps indicate model focus regions and are not equivalent to lesion segmentation masks.